

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA0308	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2024	Date:	25-08-26

Code of Conduct

I M Poojitha Reddy - 192472352 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M Poojitha Reddy

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. AI-Powered Insurance Claim Processing System

An insurance company is developing an NLP-based system to automatically process customer insurance claims. The system uses **Context-Free Grammar (CFG)** to parse customer statements and extract information such as the claimant, damaged object, cause of damage, and claim amount.

However, the system faces the following problems:

- Ambiguous sentence structures
- Difficulty distinguishing noun and prepositional phrase attachment
- Subject–verb agreement errors
- Inefficient parsing of lengthy customer descriptions
- Difficulty handling conversational and incomplete input

Example customer statement:

"I reported the damage to the car after the accident."

The sentence can produce different interpretations depending on whether "**to the car**" is associated with *reported* or *damage*.

Tasks:

- a) Analyze the structural ambiguity in the given sentence using CFG and illustrate the possible interpretations.
- b) Evaluate the limitations of a basic CFG in handling ambiguity, agreement, and long customer-generated insurance statements.
- c) Design an improved parsing approach using **PCFG, Feature Structures, and Earley Parsing** to resolve ambiguity and efficiently process the claim.
- d) Justify how the proposed architecture can improve the accuracy, scalability, and reliability of automated insurance claim processing.

2. Intelligent Railway Ticket Booking Assistant

A railway company is developing a conversational AI assistant for ticket booking. The system currently uses **top-down parsing** to interpret passenger requests.

The system encounters:

- Excessive backtracking
- Ambiguous attachment of phrases
- Incomplete passenger requests
- Long-distance dependencies
- Slow response for complex queries

Example passenger request:

"Book two tickets from Chennai to Delhi for my parents with AC sleeper."

The system sometimes incorrectly associates "**with AC sleeper**" with the passengers instead of the ticket/class information.

Tasks:

- a) Analyze the possible syntactic interpretations of the given passenger request using CFG.
- b) Evaluate why top-down parsing may result in excessive backtracking and inefficient processing for conversational ticket-booking queries.
- c) Analyze how **Earley Parsing** can handle ambiguity, incomplete input, and different possible parse structures more effectively.

d) Compare **Top-Down Parsing, CFG-based parsing, and Earley Parsing** in terms of ambiguity handling, computational efficiency, partial-input processing, and suitability for real-time railway booking systems.

3. AI-Based Legal Contract Analysis System

A legal technology company is developing an NLP system to analyze contracts and automatically identify:

- Parties involved
- Obligations
- Conditions
- Deadlines
- Actions
- Exceptions

The existing system uses a basic CFG parser. However, it produces incorrect interpretations when processing long legal sentences containing relative clauses, passive constructions, and multiple prepositional phrases.

Example contract sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

The system sometimes incorrectly identifies the **company** as the entity responsible for delivering the equipment.

Tasks:

a) Analyze the syntactic structure of the given sentence using CFG and identify possible sources of ambiguity.

b) Evaluate why basic CFG is insufficient for accurately interpreting complex legal sentences containing relative clauses and nested structures.

c) Design an NLP architecture integrating **CFG, PCFG, Feature Structures, and Earley Parsing** to improve syntactic and semantic interpretation.

d) Develop a step-by-step processing workflow showing how the system can extract:

- Supplier
- Action
- Equipment
- Company
- Obligation
- Deadline

and justify how the proposed approach improves automated contract analysis.

4. AI-Based E-Commerce Product Search System

An e-commerce company is developing a conversational product-search system. Customers enter natural-language queries rather than selecting predefined filters.

The current system uses CFG but faces difficulties with:

- Ambiguous phrase attachment
- Coordination
- Missing words
- Informal customer queries
- Long product descriptions
- Real-time search requirements

Example user query:

"Find laptops with a touchscreen for students under ₹60,000."

The system may incorrectly associate "**for students**" with the touchscreen rather than the laptop.

Tasks:

- a) Analyze the syntactic ambiguity in the given query and construct possible parse interpretations using CFG.
- b) Evaluate the limitations of CFG in handling informal, elliptical, and ambiguous e-commerce search queries.
- c) Propose an improved parsing architecture using **PCFG, Feature Structures, and Earley Parsing** to identify product attributes, user requirements, and price constraints.
- d) Justify how the proposed solution can improve search precision, response time, and customer experience in a large-scale e-commerce platform.

5. Intelligent Airline Voice Assistant

An airline is developing a real-time voice assistant that allows passengers to modify flight bookings using spoken commands.

The system currently uses a conventional top-down parser. It experiences:

- Backtracking
- Ambiguous attachment
- Speech recognition errors
- Incomplete commands
- Delayed responses
- Difficulty interpreting corrections made by users

Example spoken command:

"Change my flight to Mumbai tomorrow with two passengers."

Depending on the parse, "**with two passengers**" may be incorrectly interpreted as an attribute of the destination rather than the number of passengers.

The passenger may also say:

"Change my flight to Mumbai tomorrow... actually, make that Bangalore."

Tasks:

- a) Analyze the possible syntactic structures and ambiguities in the given voice command.
- b) Evaluate the limitations of top-down parsing when processing incomplete, corrected, and ambiguous spoken commands in real time.
- c) Design a parsing architecture using **Earley Parsing, PCFG, and Feature Structures** to support partial input, ambiguity resolution, and agreement constraints.
- d) Develop a processing workflow showing how speech input can be converted into a structured booking request and justify the proposed method for real-time airline applications.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1	4	4	4	2	3	2	19	95
2	4	4	4	2	3	2	19	
3	4	4	4	2	3	2	19	
4	4	4	4	2	3	2	19	
5	4	4	4	2	3	2	19	

Faculty In-charge	Course Coordinator	Program Director
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Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

UNIT – PARSING AND GRAMMAR

1. AI-Powered Insurance Claim Processing System

a) Analyze the structural ambiguity in the given sentence using CFG and illustrate the possible interpretations.

Answer:

The sentence is:

“I reported the damage to the car after the accident.”

The ambiguity occurs because the prepositional phrase **“to the car”** can be attached to different parts of the sentence.

A simplified CFG can be represented as:

$S \rightarrow NP VP$

$VP \rightarrow V NP PP$

$NP \rightarrow Det N \mid Pronoun$

$PP \rightarrow P NP$

Interpretation 1: “to the car” modifies “damage”

The meaning is that the **damage was to the car**.

[I] [reported [the damage [to the car]] [after the accident]]

Here:

- **I** = Claimant
- **damage** = Damaged object/event
- **car** = Damaged object
- **after the accident** = Time/Cause context

Interpretation 2: “to the car” modifies “reported”

The meaning can be interpreted as the speaker **reported the damage to someone associated with the car**, although this reading is less natural.

[I] [reported [the damage] [to the car]] [after the accident]

Thus, the sentence has a **prepositional phrase attachment ambiguity**. A basic CFG can generate both structures but cannot easily determine which interpretation is intended.

b) Evaluate the limitations of a basic CFG.

Answer:

A basic CFG describes valid grammatical structures using production rules, but it has several limitations for insurance claim processing.

1. **Ambiguity:** CFG can generate multiple valid parse trees without determining the most appropriate interpretation.
2. **Agreement:** Basic CFG does not naturally enforce subject–verb agreement such as “*The customer reports*” versus “*The customers report.*”
3. **Long sentences:** Large customer descriptions may produce many possible parse trees, increasing computational cost.
4. **Conversational input:** Incomplete sentences such as “*Car damaged yesterday, need claim*” may not fit standard CFG rules.
5. **Attachment problems:** CFG may not reliably determine whether a prepositional phrase modifies a noun or a verb.

Therefore, basic CFG alone is insufficient for robust real-world insurance claim processing.

c) Design an improved parsing approach using PCFG, Feature Structures, and Earley Parsing.

Answer:

An improved system can combine three techniques.

1. PCFG:

A Probabilistic Context-Free Grammar assigns probabilities to different parse structures. The most probable interpretation can therefore be selected when multiple parses are possible.

2. Feature Structures:

Feature structures can represent grammatical information such as **number, person, tense, and**

agreement. They can also represent insurance-specific information such as claimant, vehicle, damage type, and claim amount.

3. Earley Parsing:

Earley Parsing is suitable for CFG parsing and can handle **ambiguous structures and incomplete input** efficiently. It can process customer descriptions even when the input is conversational or partially formed.

The architecture can therefore be:

Customer Input → Preprocessing → Earley Parser → PCFG Ranking → Feature Validation → Semantic Role Extraction → Claim Processing

d) Justify how the proposed architecture improves claim processing.

Answer:

The proposed architecture improves **accuracy** by using PCFG probabilities to select the most likely interpretation and Feature Structures to enforce grammatical and semantic constraints.

It improves **scalability** because Earley Parsing can efficiently process different sentence structures, including incomplete input. The system can also be extended with insurance-specific grammar rules.

Finally, it improves **reliability** by validating extracted information before processing the claim. This reduces errors in identifying the claimant, damaged object, cause, and claim amount, resulting in more dependable automated insurance processing.

2. Intelligent Railway Ticket Booking Assistant

a) Analyze the possible syntactic interpretations using CFG.

Answer:

The request is:

“Book two tickets from Chennai to Delhi for my parents with AC sleeper.”

A simplified CFG can be represented as:

$S \rightarrow VP$

$VP \rightarrow V\ NP\ PP\ PP\ PP$

$NP \rightarrow Det\ Num\ N$

$PP \rightarrow P\ NP$

The sentence contains several prepositional phrases:

- **from Chennai** → Source
- **to Delhi** → Destination
- **for my parents** → Passengers/Travellers
- **with AC sleeper** → Ticket class/type

The intended structure is:

[Book [two tickets] [from Chennai] [to Delhi] [for my parents] [with AC sleeper]]

Here, **“with AC sleeper”** should modify the **tickets**, indicating the desired class or accommodation.

However, a parser could incorrectly associate it with **“my parents”**, producing an unintended interpretation.

b) Evaluate why top-down parsing may result in excessive backtracking.

Answer:

Top-down parsing begins with the start symbol and attempts to predict the grammatical structure of the sentence. When several grammar rules are possible, it may choose one and later discover that the choice does not match the remaining input.

This results in **backtracking**, where the parser returns to an earlier decision and tries another alternative. Conversational ticket requests contain multiple prepositional phrases, optional words, and ambiguous attachments, which can significantly increase backtracking.

For long or complex requests, this can increase processing time and reduce the responsiveness required by a real-time railway booking assistant.

c) Analyze how Earley Parsing handles ambiguity and incomplete input.

Answer:

Earley Parsing is a dynamic-programming parsing technique that maintains partial parsing states while processing the input. It uses three important operations:

- **Prediction** – predicts possible grammar rules.
- **Scanning** – matches grammar symbols with input words.
- **Completion** – completes partially recognized rules.

It can maintain multiple possible interpretations without repeatedly restarting the entire parse. It can also process **partial or incomplete input**, which is particularly useful in conversational systems.

Therefore, Earley Parsing is more suitable for railway queries where users may provide information gradually, such as “**Book two tickets... from Chennai...**” before completing the request.

d) Compare Top-Down, CFG-based, and Earley Parsing.

Answer:

Feature	Top-Down Parsing	BasicCFG Parsing	Earley Parsing
Ambiguity	May require backtracking	Can generate multiple parses	Handles multiple parses systematically
Efficiency	Can become inefficient due to backtracking	Depends on parsing method	Efficient dynamic programming approach
Partial Input	Generally difficult	Limited	Supports incomplete/partial structures
Long Dependencies	Can be difficult	Depends on grammar	Better suited to complex structures
Real-Time Use	May become slow	Moderate	More suitable
Conversational Queries	Limited	Requires extensive grammar	More flexible

Therefore, **Earley Parsing is generally the better choice** for a real-time railway booking assistant because it can handle ambiguity, incomplete input, and complex structures more effectively.

3. AI-Based Legal Contract Analysis System

a) Analyze the syntactic structure using CFG and identify possible sources of ambiguity.

Answer:

The sentence is:

“The supplier who delivered the equipment to the company must replace the defective components within thirty days.”

A simplified CFG structure is:

$S \rightarrow NP VP$

$NP \rightarrow Det N RC$

$RC \rightarrow RelPron VP$

$VP \rightarrow V NP PP$

The major structure is:

[The supplier [who delivered [the equipment] [to the company]]] [must replace [the defective components] [within thirty days]]

The semantic roles are:

- **Supplier** → Party/Agent
- **who** → Refers to supplier
- **delivered** → Action
- **equipment** → Object
- **company** → Recipient
- **replace** → Obligation action
- **defective components** → Object
- **within thirty days** → Deadline

The main difficulty comes from the **relative clause “who delivered the equipment to the company”** and its nested prepositional phrase.

b) Evaluate why basic CFG is insufficient for complex legal sentences.

Answer:

Basic CFG can represent the grammatical structure of a sentence, but it does not contain enough information to determine semantic relationships.

Legal sentences frequently contain:

- Relative clauses
- Passive constructions
- Nested clauses
- Multiple prepositional phrases
- Long-distance dependencies
- Complex obligations and exceptions

In this example, the system may incorrectly associate “**who delivered**” with the company rather than the supplier. A basic CFG can generate the syntactic structure but cannot reliably determine the semantic role of every entity.

Therefore, legal NLP requires grammar, probabilities, grammatical features, and semantic analysis together.

c) Design an architecture integrating CFG, PCFG, Feature Structures, and Earley Parsing.

Answer:

A suitable architecture can be designed as follows:

Legal Document → Preprocessing → CFG Grammar → Earley Parser → PCFG Ranking → Feature Structure Validation → Semantic Role Labeling → Legal Information Extraction

Role of each component:

CFG:

Defines the basic grammatical structures of legal sentences.

Earley

Efficiently handles complex, ambiguous, and nested grammatical structures.

Parser:

PCFG:

Assigns probabilities to alternative parse trees and selects the most likely interpretation.

Feature

Store grammatical features such as number, person, tense, voice, and agreement. They can also maintain legal semantic information such as party and obligation roles.

Structures:

This combination provides both syntactic and semantic information needed for accurate contract analysis.

d) Develop a step-by-step workflow for extracting the required information.

Answer:

Step 1 – Preprocessing

The system tokenizes the contract sentence and identifies important entities such as **supplier, equipment, company, components, and thirty days**.

Step 2 – Syntactic Parsing

The CFG and Earley Parser identify the main clause and relative clause:

Main**clause:**

“The supplier must replace the defective components.”

Relative**clause:**

“who delivered the equipment to the company.”

Step 3 – PCFG Disambiguation

PCFG assigns probabilities to alternative parse structures and selects the most probable syntactic interpretation.

Step 4 – Feature Validation

Feature Structures verify grammatical relationships and connect “**who**” to its antecedent, **supplier**.

Step 5 – Semantic Extraction

The system extracts:

Information Extracted Value

Supplier Supplier

Action Delivered / Replace

Equipment Equipment

Information Extracted Value

Company Recipient

Obligation Replace defective components

Deadline Within 30 days

Step 6 – Legal Interpretation

The system concludes that **the supplier has an obligation to replace defective components within thirty days.**

This architecture improves contract analysis by correctly connecting entities with their actions and distinguishing **obligations, recipients, objects, and deadlines.**

4. AI-Based E-Commerce Product Search System

a) Analyze the syntactic ambiguity and construct possible CFG interpretations.

Answer:

The query is:

“Find laptops with a touchscreen for students under ₹60,000.”

A simplified CFG can be represented as:

$S \rightarrow VP$

$VP \rightarrow V NP$

$NP \rightarrow N PP PP PP$

$PP \rightarrow P NP$

The intended interpretation is:

Find [laptops [with a touchscreen] [for students] [under ₹60,000]].

Here:

- **Laptops** → Product

- **with a touchscreen** → Product attribute
- **for students** → Target user
- **under ₹60,000** → Price constraint

An incorrect parse could associate “**for students**” directly with “**touchscreen**”, suggesting that the touchscreen is specifically for students rather than describing the target product.

Thus, the main ambiguity is **prepositional phrase attachment**.

b) Evaluate the limitations of CFG for informal e-commerce queries.

Answer:

Basic CFG has difficulty handling natural e-commerce queries because customers frequently use **informal, incomplete, and abbreviated language**.

For example:

- “Laptop under 60k”
- “Good gaming laptop?”
- “Phones with best camera”
- “Need laptop for college”

Such queries may not follow complete grammatical sentence structures.

CFG also has difficulty with **phrase attachment, coordination, missing words, and long queries**. It may produce multiple parse trees without knowing which interpretation best matches the customer's search intention.

Therefore, basic CFG alone is not sufficient for modern conversational product search.

c) Propose an improved parsing architecture.

Answer:

The proposed architecture is:

User Query → Preprocessing → Earley Parsing → PCFG Ranking → Feature Structures → Product Attribute Extraction → Search Engine

PCFG

PCFG assigns probabilities to possible parse structures and helps select the most likely interpretation.

Feature Structures

Feature structures represent product information such as:

- Product = Laptop
- Screen = Touchscreen
- Target User = Student
- Price = ₹60,000 maximum

They can also enforce constraints and agreement relationships.

Earley Parsing

Earley Parsing efficiently handles ambiguous and incomplete queries and maintains alternative interpretations during parsing.

The final structured query becomes:

Product = Laptop

Feature = Touchscreen

TargetUser = Student

Price ≤ ₹60,000

This structured representation can then be sent to the product-search engine.

d) Justify how the proposed solution improves search performance.

Answer:

The proposed architecture improves **search precision** by accurately separating product attributes, user requirements, and price constraints.

It improves **response time** because Earley Parsing efficiently handles complex and incomplete queries without excessive backtracking. PCFG helps prioritize the most probable interpretation instead of processing every possible interpretation equally.

It also improves **customer experience** because users can express their requirements naturally instead of selecting multiple predefined filters. The system can understand queries such as “**laptops with touchscreen for students under ₹60,000**” and convert them into accurate search constraints.

5. Intelligent Airline Voice Assistant

a) Analyze the possible syntactic structures and ambiguities.

Answer:

The command is:

“Change my flight to Mumbai tomorrow with two passengers.”

A possible intended structure is:

[Change [my flight] [to Mumbai] [tomorrow] [with two passengers]]

The semantic interpretation is:

- **Action:** Change
- **Object:** Flight
- **Destination:** Mumbai
- **Date:** Tomorrow
- **Passenger Count:** Two

The ambiguity occurs with **“with two passengers.”** It could potentially be attached to the destination phrase or interpreted as information about the flight.

The intended interpretation is that **two passengers are travelling on the modified flight.**

The correction:

“Change my flight to Mumbai tomorrow... actually, make that Bangalore.”

creates another challenge. The system must understand that **Bangalore replaces Mumbai** rather than treating both Mumbai and Bangalore as destinations.

b) Evaluate the limitations of top-down parsing for spoken commands.

Answer:

Top-down parsing can suffer from excessive **backtracking** when several grammatical structures are possible. This is problematic for voice assistants because users often speak in incomplete or corrected sentences.

Speech recognition can also produce errors, and users may change their minds during a sentence. For example:

“Mumbai... actually, make that Bangalore.”

A conventional parser may initially create a structure containing Mumbai and then have difficulty revising it.

Therefore, top-down parsing may produce **delays, incorrect interpretations, and poor handling of conversational corrections**, which are undesirable in real-time airline systems.

c) Design a parsing architecture using Earley Parsing, PCFG, and Feature Structures.

Answer:

A suitable architecture is:

Speech Input → Speech Recognition → Text Normalization → Earley Parser → PCFG Ranking → Feature Structure Validation → Semantic Interpretation → Booking System

Earley Parsing

Earley Parsing maintains partial parsing states and can handle **incomplete commands and multiple possible structures**.

PCFG

PCFG assigns probabilities to alternative interpretations and selects the most likely parse.

For example, it can assign a higher probability to:

`with two passengers → PassengerCount`

rather than treating it as a destination attribute.

Feature Structures

Feature Structures maintain structured information such as:

`Destination = Mumbai`

`Date = Tomorrow`

`PassengerCount = 2`

When the user says:

“Actually, make that Bangalore.”

the feature:

`Destination = Mumbai`

is updated to:

`Destination = Bangalore`

Thus, the system can dynamically revise the booking request.

d) Develop a processing workflow and justify the proposed method.

Answer:

Step 1 – Speech Recognition

The passenger's voice is converted into text:

“Change my flight to Mumbai tomorrow with two passengers.”

Step 2 – Syntactic Parsing

Earley Parsing identifies the possible grammatical structures and maintains partial parses.

Step 3 – Ambiguity Resolution

PCFG ranks the possible interpretations and selects the most probable structure.

Step 4 – Feature Extraction

Feature Structures create a structured booking request:

Feature	Value
Action	Change Flight
Destination	Mumbai
Date	Tomorrow
Passengers	2

Step 5 – Handle Correction

When the passenger says:

“Actually, make that Bangalore.”

the system updates:

Destination: Mumbai → Bangalore

Step 6 – Validation

The system checks that the destination, date, passenger count, and other booking details are valid.

Step 7 – Booking Execution

The validated structured request is passed to the airline booking system.

Conclusion

The combination of **Earley Parsing, PCFG, and Feature Structures** provides better handling of ambiguity, partial speech, corrections, and grammatical constraints. It therefore provides a more **accurate, responsive, and reliable solution for real-time airline voice assistants**.